Experiments with microwaves

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Abstract

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1 Introduction

The spectrum of electromagnetic radiation covers a wide range of wavelengths, ranging from high-energy γ-rays to low-energy radio waves, all of which play an important role in today's world. Examples include X-rays used in medical diagnostics, the radio waves employed by mobile phones to transmit information, and microwaves used to heat your lunch.
In this experiment, we take a closer look at the latter type of radiation: microwaves. Their wavelengths range from one meter to one millimeter, giving them a wide variety of applications. Furthermore, microwaves play a significant role in our understanding of the universe through the study of the cosmic microwave background, which is believed to be a remnant of the Big Bang, the event that marked the beginning of the universe. Microwaves exhibit many interesting properties. In the following, several of these properties will be examined and discussed, namely the angular distribution and polarization properties of microwaves, their wavelengths, and the formation of evanescent waves. Using these phenomena, we will also determine the lattice constant of a macroscopic cube.

2 Angular distribution of microwaves

To examine the angular distribution of microwaves, the transmitter was placed on a goniometertable, such that the horn is located approximately at the pivot point. The angular distribution was measured to both sides $\pm 90°$ in equidistand steps. The measurements are done with a photo diode that is connected to an Agilent 34402A. The measured intensity was normalized to 1 for $\theta = 0°$ and is shown in fig. 1. The density of measured points around $\theta = 0°$ is quite low, which is not ideal. One should see the symmetry around the 0-point, but with our measurements, one can only guess it.

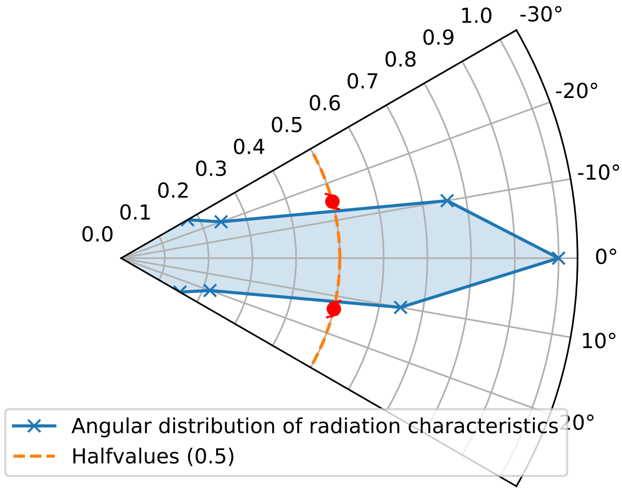

Figure 1: Measured voltage within the relevant angular range normalized to 1 in dependence of the angle θ. The left and right full width at half maximum are marked.

Nevertheless was the angle stretch of the radiation quantified by calculating the halfangle θ_H where the full width at half maximum lies.

$$\theta_H = (28.4 \pm 2.5)° \tag{1}$$

As expected due to the transmitter being a horn antenna, this confirms that the microwave radiation is pointed towards the 0° direction.

3 Focal lenght of paraffin wax lens

To further investigate microwaves, it is necessary to produce collimated rays. For that a paraffin wax lens was used which focal point has to be determined. This was archieved by placing a wax lens between transmitter and receiver. The distance between the transmitter and the las is $g = (80 \pm 1)cm$. The re-

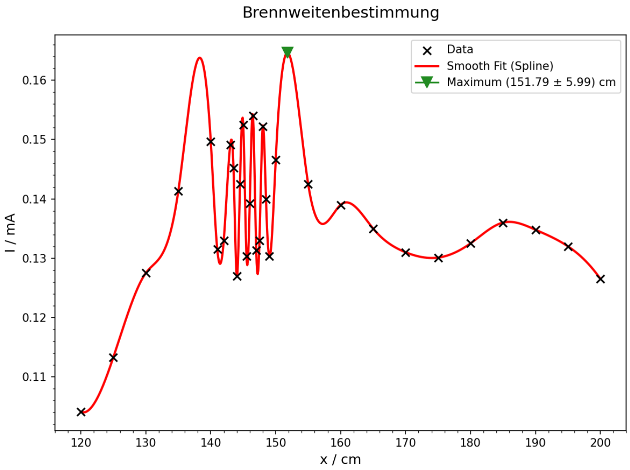

Figure 2: Intensity profile of the microwave radiation in relation to it's distance from the lens.

ceiver is placed with maximal possible distance to the lense and then progressively moved towards it. The radiation was measured in every step. To determine the peak radiation level, a smooth curve was fitted to the experimental data and it's maximum was determined to $b' = (151.8 \pm 6.0) cm$. The plot is shown in fig. 2.
The focal length was then calculated using the thin-lense equation:

$$f = \frac{g \cdot b}{g + b} \quad \text{with} \quad b = b' - g \tag{2}$$

which results in a focal lenght of $(f = 37.84 \pm 0.48(cm)$. In the following parts of the experiment, the focal lenght was roughly placed at 40 cm, such that the rays would be collimated.

4 Determination of the lattice constant of a cubic crystal

In solid state physics, the θ-rocking-curve method is often used to examine the characteristics of an atomic lattice, namly the lattice constant. In this part of the experiment, this method was tested with microwaves on a macroscopic styrofoam cube doted with aluminum wrapped balls, such that a cubic lattice emerges. The bragg's law states that monochromatic radiation of wavelenght λ interferes constructive under the angle θ if

$$n\lambda = 2dsin(\theta) \tag{3}$$

is realised. Here n is the number of the maximum or diffraction order, λ the wavelenght measured in a previous part, d the spacing between adjacent crystal planes, and θ is the angle at which the maximum occurs. The setup used in the experiment is shown in

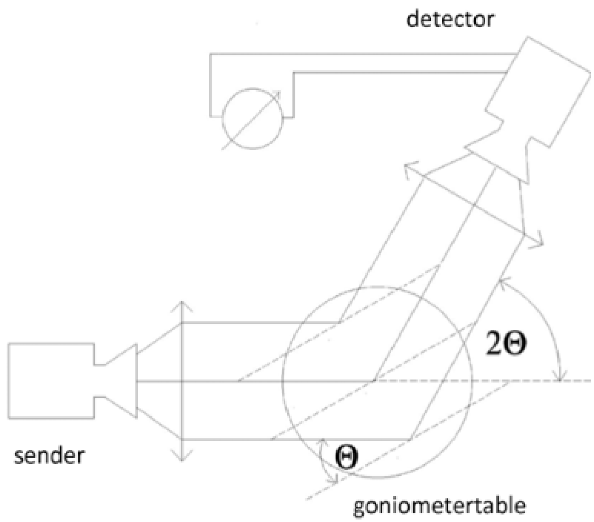

Figure 3: Experimental setup to measure the lattice constant of an macroscopic cube

fig. 3. Starting the measurement at an angle $\theta = 0°$, the crystal was rotated in steps of $2°$ while the detector was rotated to be at 2θ to ensure the symmetric condition required for Bragg reflection.
While Bragg's law is independent of the orientation of the crystal planes, it acutally plays a role in the measurement. That is why we need to introduce Miller indices h, k, l to describe which plane we are currently measuring. For a cubic crystal, the spacing between lattice plane can be expressed as:

$$d_{hkl} = \frac{d_0}{\sqrt{h^2 + k^2 + l^2}} \tag{4}$$

Using equations 3 and 4 one can calculate the lattice constant d_0 with:

$$d_0 = d_{hkl}\sqrt{h^2 + k^2 + l^2} = \frac{n\lambda}{2sin(\theta)}\sqrt{h^2 + k^2 + l^2} \tag{5}$$

The sides with the Miller indicies (100) and (110) were measured and the intensity in terms of the rotation angle is shown in fig. 4 and 5 respectively.

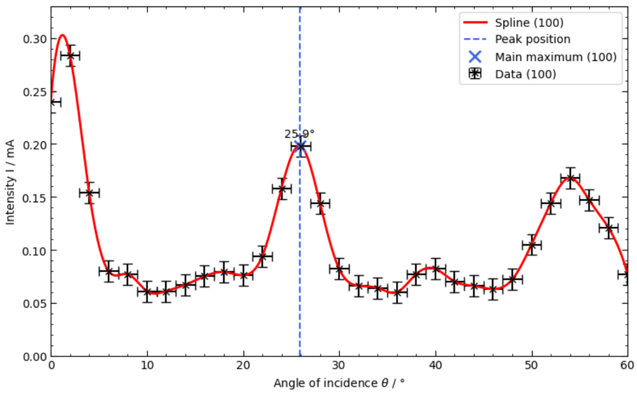

Figure 4: Measured Bragg diffraction from the (100) plane for angles $\theta = [0, 60]$. The first maximum was determined and marked.

The first maximum for the (100) plane was measured at an angle $\theta = (25.89 \pm 0.5)°$.

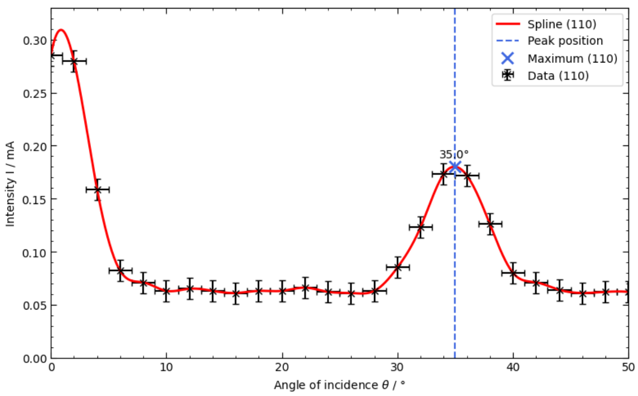

Figure 5: Measured Bragg diffraction from the (110) plane for angles $\theta = [0, 50]$. The first maximum was determined and marked.

The first maximum for the (110) plane was measured at an angle $\theta = (34.95 \pm 0.5)°$.
Using eq. 5 and the wavelenght $\lambda = hehrher$ the lattice constant can be calculated:

$$\begin{aligned} (100) : d_0 &= (3.74 \pm 0.14) \text{ cm} \\ (110) : d_0 &= (4.03 \pm 0.10) \text{ cm} \end{aligned} \tag{6}$$

Both values for the lattice constant are in a reasonable order of magnitude considering the dimension of the styrofoam cube, but aren't consistent within their uncertainties. The deviation could have it's origin in the 80's style experimental setup where the cube was loosely placed on the goniometric table and therefore not perfectly aligned.
None the less this experiment shows that this method of examining an unknown lattice structure is working and with numerous alternations of the setup e.g. a more modern type, it can yield reliable data.

5 Properties of a circular waveguid

Waveguides are resonators with two open ends, allowing waves to propagate on its interior from one end to the other with minimal loss of energy by restricting the transmission of energy to only the direction of the guide. There are three types of waveguides: acoustic waveguides which direct sound, optical waveguides which direct light, and radiofrequency waveguides which direct electromagnetic waves other than visible or near visible light.
In this part of the experiment one of the last type was used to guide microwaves. It is a hollow conductive metal pipe which inner walls are often plated with a material which reflects signals as much as possible such that the loss is reduced. For high end waveguides silver or gold is often used. It works by trapping and "bouncing" the waves off its highly conductive inner walls, directing the energy forward. With that there can be two different propagation modes, one where the electric field is completely perpendicular to the direction the wave is traveling and one where the magnetic field is completely perpendicular to the travel direction of the wave.
To investigate this a transmitter was placed at one, a receiver was placed at the other end of a waveguide. Despite the curvature and lenght of the guide a clear signal was detected. To verify that the the signal was transmitted through the interior of the pipe and not just on its surface, we closed one end with a metal cap. Since no signal, besides the background noise could be detected, it is proof to the previous claim.
Finally we examined the polarization of the transmitted waves by rotating the receiver. A strong dependence of the measured signal to the rotation of the receiver was observed, signaling that the wave leaving the waveguide is indeed polarized. The maximum was observed at a rotation angle of approximately 90° relative to the transmitter. This indicates that the polarization direction was altered during propagation through the waveguide.

References

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